Internet Crimes and Cyber Security

Week: 06 **Session: 30**

Topic :Introduction to Cyber Threat Intelligence

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Understanding Cyber Threat Intelligence

Concepts to be covered

1. What is Cyber Threat Intelligence
2. Understanding Cyber Threat Intelligence Life Cycle
3. Cyber Threat Intelligence Processing and Exploitations
4. Cyber Threat Intelligence Platforms
5. Live Case Study : Cyber Threat Intelligence Evaluation
6. Conclusion and Way Ahead

Understanding Cyber Threat Intelligence

1. What is Cyber Threat Intelligence

Cyber Threat Intelligence refers to the collection, processing, and analysis of data to understand the motives, targets, and methods of threat actors. It transforms raw data into actionable insights, enabling security teams to make informed, data-driven decisions. This proactive approach helps organizations anticipate and mitigate potential cyber threats before they materialize.

Key Characteristics of Cyber Threat Intelligence

Organization-Specific: CTI focuses on the unique vulnerabilities and threats specific to an organization, rather than general information about hypothetical threats.

Detailed and Contextual: It provides insights into the tactics, techniques, and procedures (TTPs) used by threat actors, along with indicators of compromise (IoCs) that signal potential attacks.

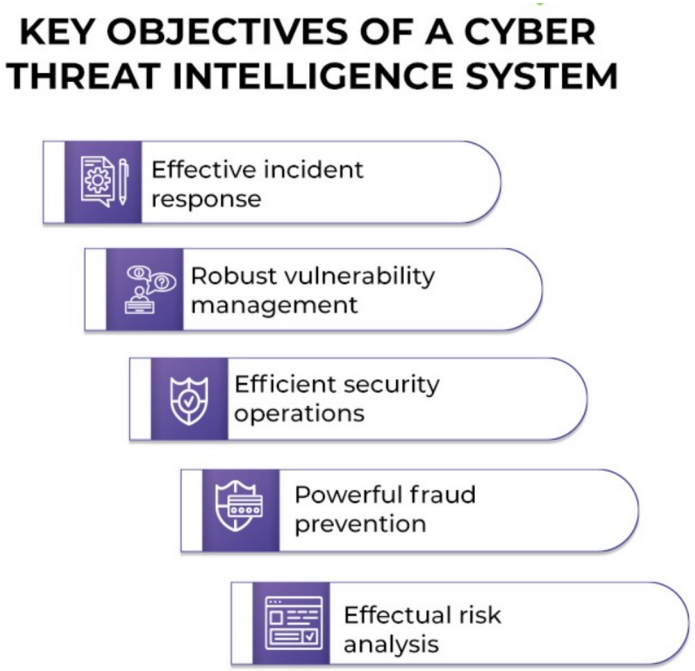
Actionable: The intelligence gathered is designed to inform security teams on how to address vulnerabilities, prioritize threats, and improve overall security posture.

Types of Cyber Threat Intelligence

Strategic Intelligence: High-level insights that inform decision-making and long-term security strategies.

Operational Intelligence: Information about specific threats and threat actors, including their methods and motivations.

Tactical Intelligence: Focused on immediate threats, providing technical details such as IoCs that can be integrated into security tools.



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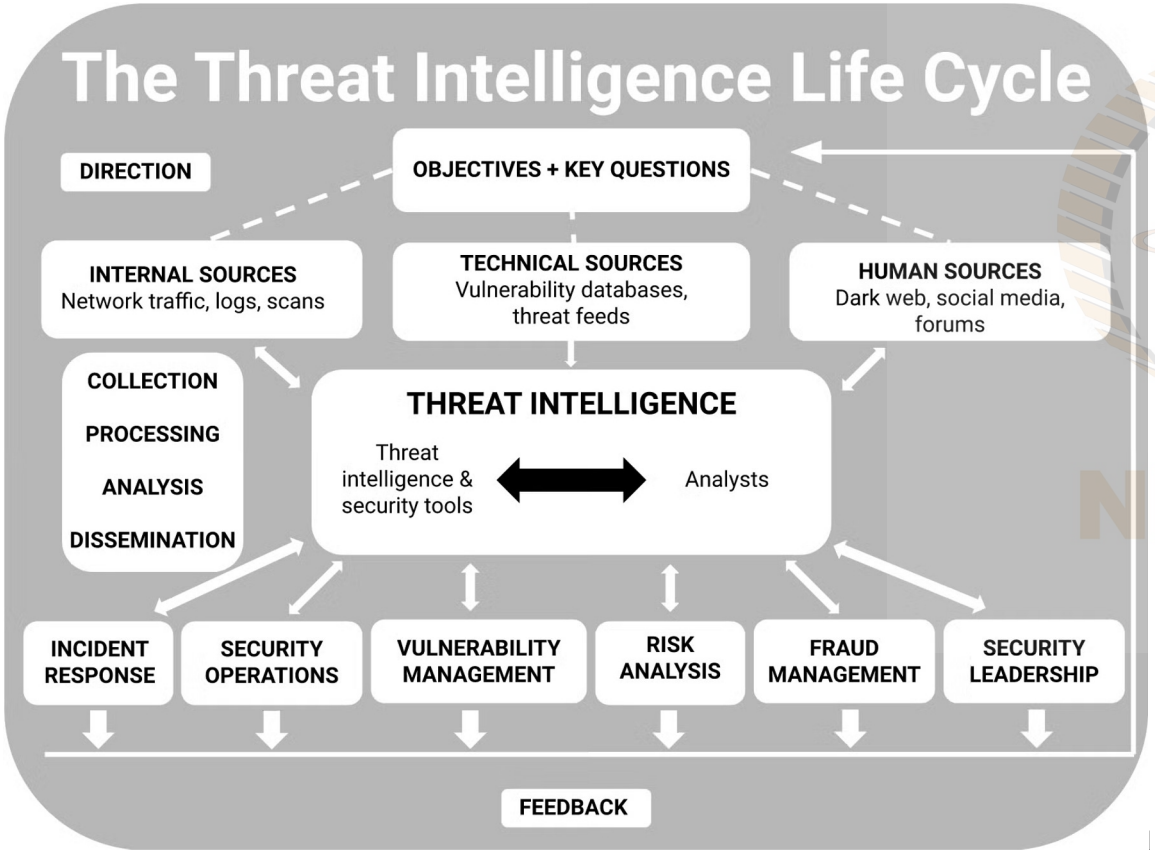
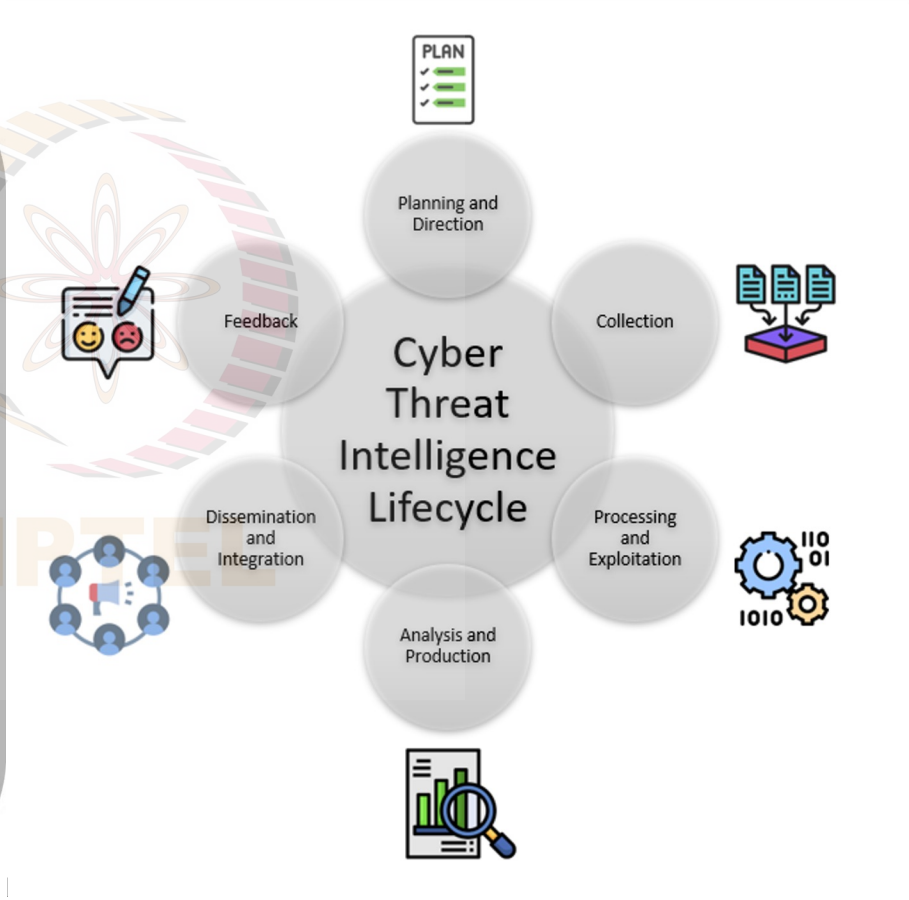
2. Understanding Cyber Threat Intelligence Life Cycle





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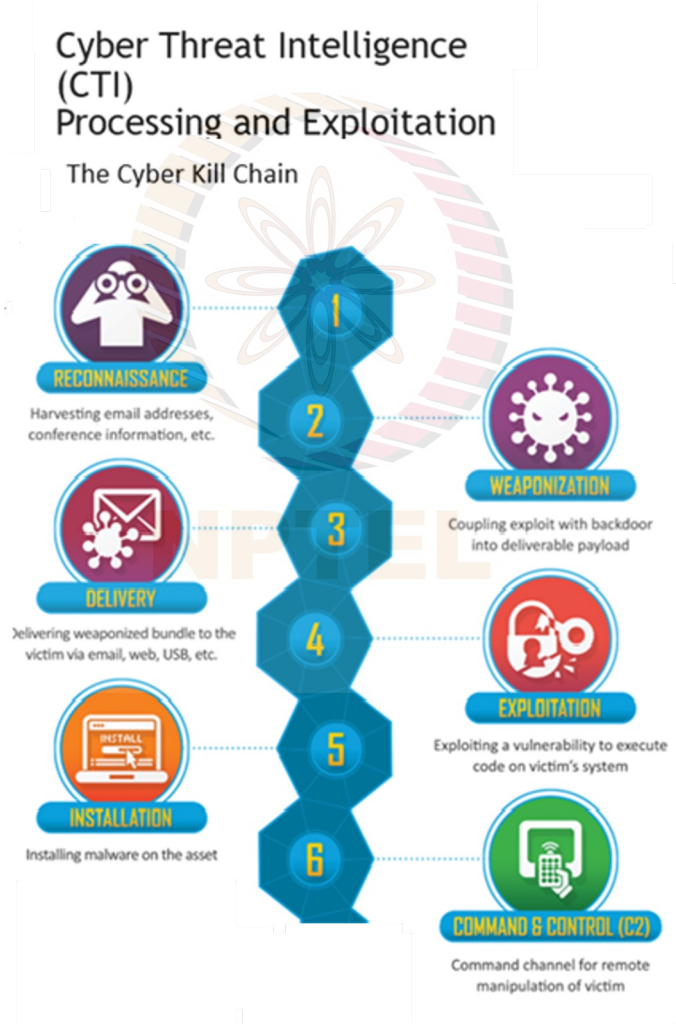
3. Cyber Threat Intelligence Processing and Exploitations

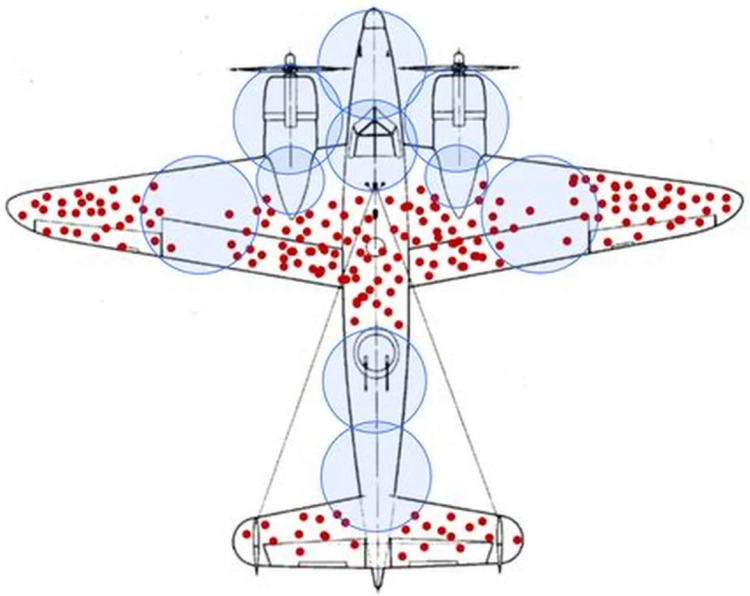
What is Cyber Kill Chain

The Cyber Kill Chain was developed by Lock heed Martin to conceptualize the step-by-step process used by attackers to infiltrate s ystems and achieve their objectives.

By mapping the attack lifecycle, organizatio ns can identify points of intervention and str engthen cybersecurity defenses. The model divides attacks into sequential stages, from initial reconnaissance to achieving the attac ker's goal.



Abraham Wald: Survivorship Bias



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4. Cyber Threat Intelligence Platforms

1. SOCRadar Free Edition: Used foe AI-powered phishing, supply chain compromises,

2. VirusTotal : Has ability to quickly scan and assess suspicious files, URLs, domains, and IPs

3. AlienVault OTX : largest community-powered threat intelligence sharing platforms, widely used by security teams, threat researchers, and SOC analysts

4. MISP (Malware Information Sharing Platform) : a widely used open-source platform for sharing, storing, and analyzing threat intelligence.

5. Cisco Talos' Intelligence Center: Cisco Talos leverages its extensive network telemetry and security product ecosystem to provide highly actionable intelligence.

6. GreyNoise: provides real-time insights into which CVEs, attack techniques, and protocols are being actively scanned, giving organizations an early warning about potential exploitation trends.

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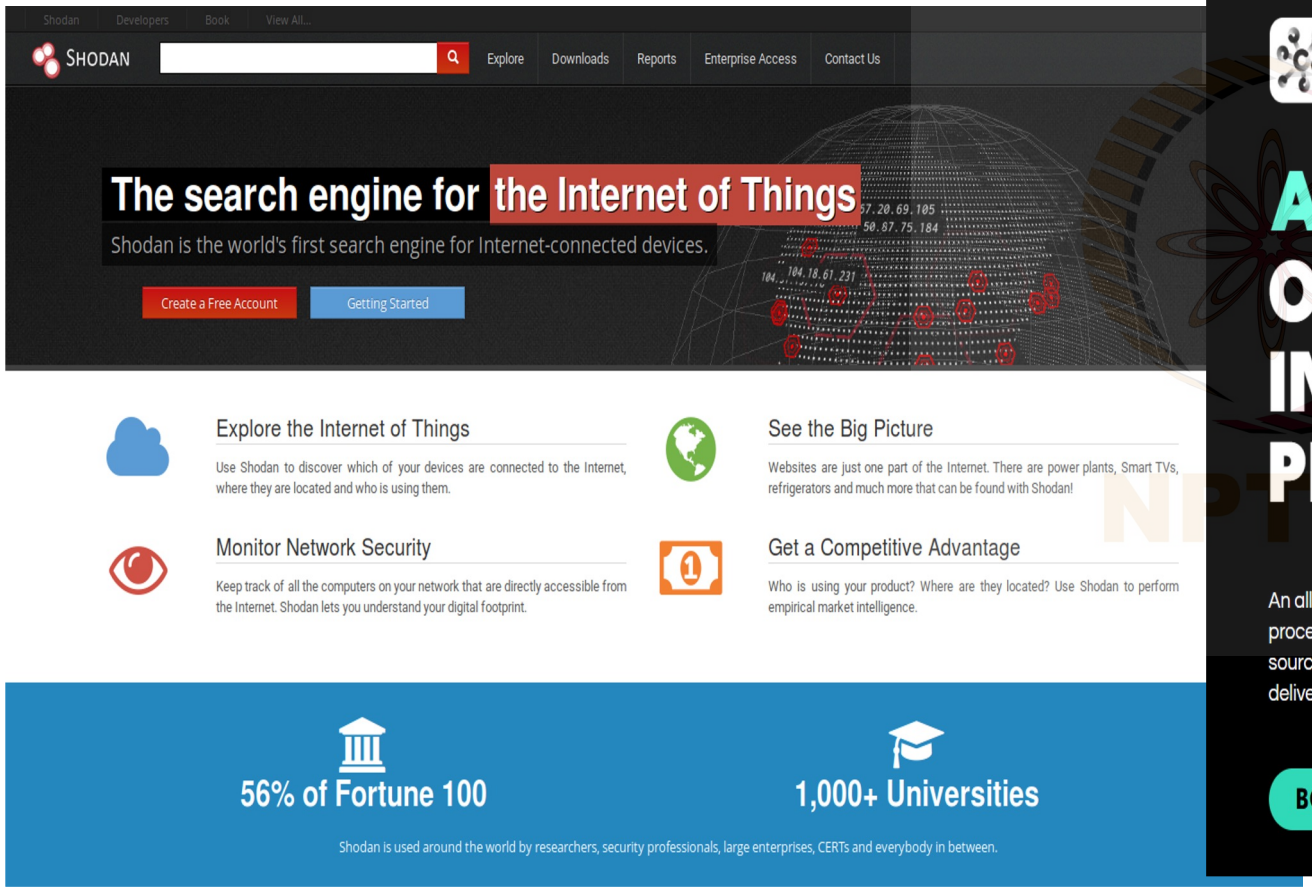
5. Live Case Study of Cyber Threat Intelligence Platform



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6. Conclusion and Way Ahead

In the realm of cyber threat intelligence (CTI), the future lies in leveraging advanced technologies like artificial intelligence (AI) and machine learning to enhance detection and response capabilities. These technologies automate data processes, improve predictive capabilities, and enable real-time threat detection, significantly improving cyber defense strategies.

Additionally, the integration of collaboration among organizations and the use of predictive analytics to forecast potential security incidents are crucial for proactive measures against emerging threats. Embracing these innovations will not only safeguard assets but also position organizations as leaders in cybersecurity resilience.

As cyber threats continue to evolve, staying ahead requires continuous learning and adaptation, focusing on understanding threat actors and their tactics.

Thank You